

Autodesk® Scaleform®

HUD

Scaleform 4.2 HUD

Flash UI

C++

AAA

Nate Mitchell, Prasad Silva

2.0

2010 7 30

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Autodesk® Scaleform® 4.2

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Autodesk Scaleform

HUD

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

(301) 446-3200

(301) 446-3199

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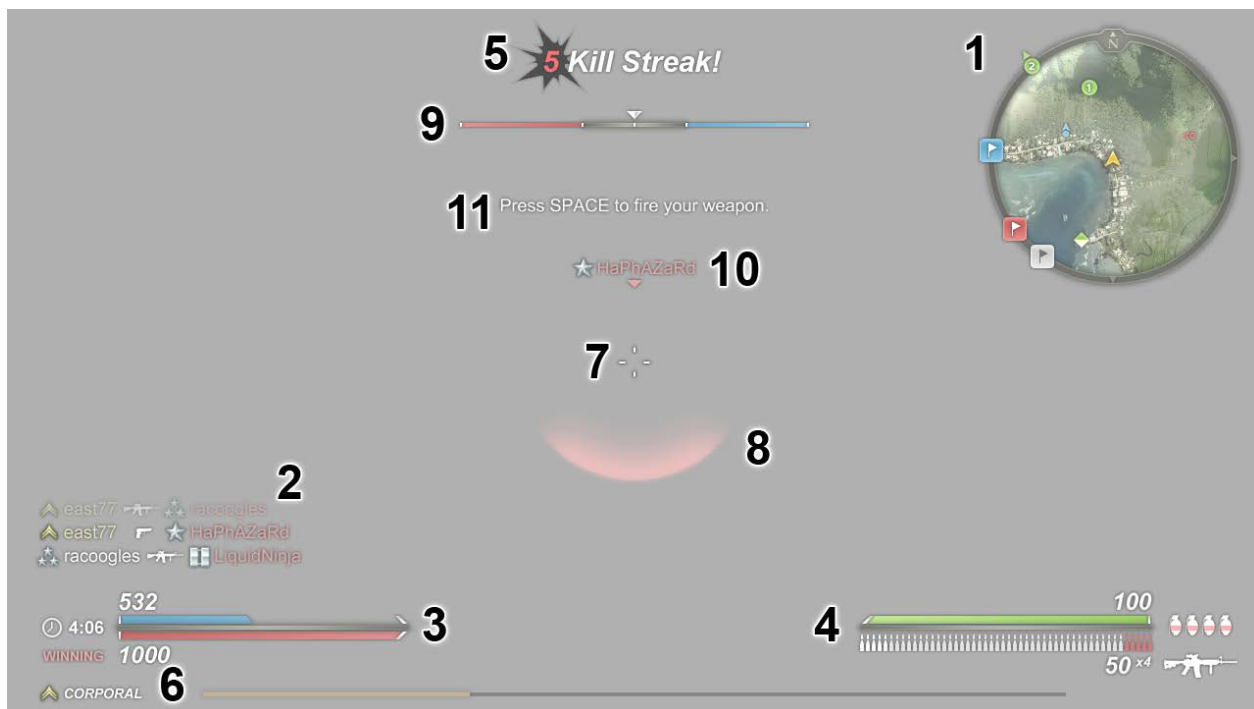
1

Scaleform HUD

Scaleform® Direct Access API

AAA

HUD
(FPS) UI



1 HUD

HUD

UI

1. (Minimap)
- 2.
- 3.
4. / /
- 5.
- 6.
- 7.
- 8.
- 9.
10. /
- 11.

FPS

2

2.1 (Build)

- *Apps\Kits\HUD* - HUD C++
- *Bin\Data\AS2 or AS3\Kits\HUD* - Flash ActionScript AS2
ActionScript 2 Flash 8 AS3 ActionScript 3 Flash 10
- *Projects\Win32 {Msvc80 or Msvc90}\Kits\HUD* - Windows Visual Studio
2005/2008
- *Projects\Xbox360 {Msvc80 or Msvc90}\Kits\HUD* - Xbox 360 Visual Studio
2005/2008
- *Projects\Common\HudKit.mk* - HUD PS3

Bin\Kits\HUD Windows HUDKit.exe
Scaleform SDK

Windows *Projects\Win32\Msvc80\Kits Msvc90\Kits* Scaleform 4.2 Kits.sln
Debugging

“Working ” *Bin\Data\AS2\Kits\HUD or Bin\Data\AS3\Kits\HUD*

Xbox 360 *Projects\Xbox360 {Msvc80 Msvc90}\Kits\HUD* Scaleform 4.2
Kits.sln

Bin\Data\AS2 or AS3\Kits\HUD Xbox 360

PS3 Scaleform *make* HUD
(HUD Kit) PS3 SN Systems .

Bin\PS3

o app_home/ Directory: *{Local Scaleform Directory}\Bin\Data\AS2 or AS3\Kits\HUD*

2.2

HUD

1

1

500

3

15

4

(Power-up)

2.3

2.3.1 (Windows)

W, A, S, D	WS	A, D
SPACE		
R		
G		
ESC	(Text Field)	
B	AI	
1, 2	• 1 • 2	

2.3.2 (Xbox/PS3)

	W, S	A, D
X /		
B /		
Y /		

3

HUD	Flash UI	HUD	C++	C++
	HUD			

3.1 C++

C++

Flash UI

Fx

HUD

MVC

HUD

Scaleform

Direct Access API

C++

Apps\Kits\HUD\

3.1.1

- **HUDKitDemo.cpp** – Scaleform Player
C++ SWF
- **HUDKitDemoConfig.** – FxPlayerConfig.h HUDKitDemo
Windows

3.1.2 HUD/

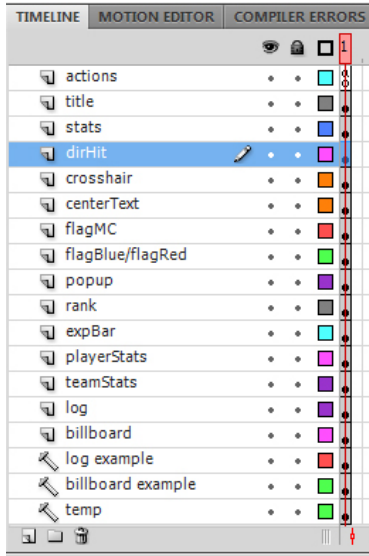
- **FxHUDKit.h** – HUD HUD
- **FxHUDView.h/.cpp** – HUD (Billboard)
- **FxMinimap.h** –
- **FxMinimapView.h/.cpp** –

3.1.3

- HUDAdapter.h/.cpp – HUD
- HUDEntityController.h/.cpp – AI
- HUDSimulation.h/.cpp – UI

3.2 *Flash*

HUD	Flash	HUDKit.fla	Minimap.fla	FLA
HUD	UI	Scaleform	UI	
Scaleform	C++	ActionScript	Flash	Flash / v



2 HUDKit.fla

HUDKitDemo	Frame 1	1	External Interface	HUDKit MovieClip
------------	---------	---	--------------------	------------------

```
ExternalInterface.call("registerHUDView", this);
```

MovieClip	C++	MovieClip	C++ HUDView
-----------	-----	-----------	-------------

“HUD”	MovieClips	Scaleform C++
-------	------------	---------------

3.2.2 Minimap.fla

3 Minimap

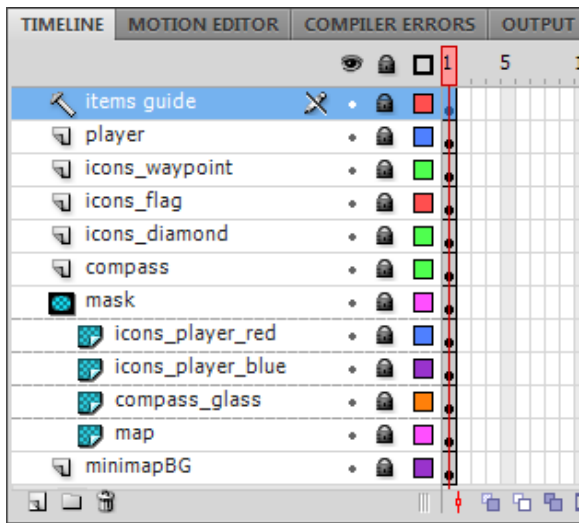
Minimap.fla 'Minimap'
3D

“ ”

-
-
- (Flag)
-

API / Direct Access
MovieClips (Stage) MovieClips C++

'Minimap'



4 'minimap'

- items guide / (Guide Layer) SWF
- player
- icons_waypoint
- icons_flag
- icons_diamond
- compass “ ”
- mask (Mask Layer)
- 0 icons_player_red

- o icons_player_blue
- o compass_glass
- o map

C++

/

- minimapBG

C++

'Minimap'

com.scaleform.Minimap

Bin\Data\AS2 or AS3\Kits\HUD\com\scaleform

HUDKitDemo

4 HUD

C++ HUD

HUDKit.swf

- FxHUDView – HUD HUDKit.swf (Core Manager)
MovieClips
- BillboardCache – /
-
- FxHUDLog – HUD
FxHUDMessages
- FxHUDMessage –

FxHUDView

MC MovieClip

FxHUDView

UpdateView()

MovieClip

HUDKit fla

UI

FxHUDView::UpdateView()

UpdateTeamStats() UpdatePlayerEXP() Update PlayerStats() UpdateEvents() UpdateTutorial()

UpdateBillboards()

UI

UpdateView()

Scaleform Direct Access API

Flash

GFx::Value

Scaleform

Flash

GFx::Value::SetDisplayInfo() GFx::Value::GetDisplayInfo()

GFx::Value::SetText()

HUD

4.1

teamStats MovieClip

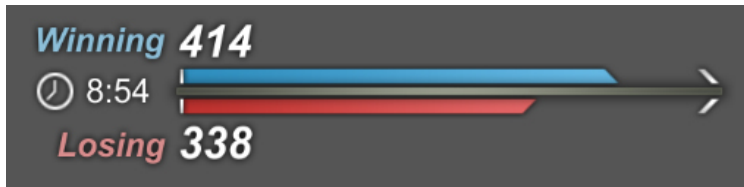
HUDKit fla

1 (Scene 1)

teamStats

(redWinning, blueWinning)
(teamRed, teamBlue)

(scoreRed, scoreBlue)
(roundTime)



5 teamStats

FxHUDView::UpdateTeamStats()

teamStats

teamStats MovieClip

```
void FxHUDView::UpdateTeamStats(FxHUDEnvironment *penv)
{
    // Retrieve the scores from the simulation.
    unsigned scoreBlue = unsigned(penv->GetHUDBlueTeamScore());
    unsigned scoreRed = unsigned(penv->GetHUDRedTeamScore());

    String text;
    Value tf;
    Value::DisplayInfo info;

    // We won't update any element who has not changed since the last update
    // to avoid unnecessary updates. To do so, we compare the previous
    // simulation State to the latest simulation data. If the two are different,
    // update the UI element in the HUD View.
    if (State.ScoreRed != scoreRed)
    {
        Format(text, "{0}", scoreRed);
        ScoreRedMC.SetText(text); Update the red team score text.

        info.SetScale((scoreRed / (Double)MaxScore) * 100, 100);
        TeamRedMC.SetDisplayInfo(info); // Update and scale the red team score
                                        bar movieClip.
    }

    if (State.ScoreBlue != scoreBlue)
    {
        Format(text, "{0}", scoreBlue);
        ScoreBlueMC.SetText(text); // Update the blue team score text.

        info.SetScale((scoreBlue / (Double)MaxScore) * 100, 100);
        TeamBlueMC.SetDisplayInfo(info); // Update and scale the blue team
                                        score bar movieClip.
    }

    if (State.ScoreRed != scoreRed || State.ScoreBlue != scoreBlue)
    {
        // Update the "Winning" and "Losing" text fields for both teams
        // if the score has changed.
        if (scoreBlue > scoreRed)
```

```

    {
        RedWinningMC.SetText(LosingText);
        BlueWinningMC.SetText(WinningText);
    }
    else
    {
        RedWinningMC.SetText(WinningText);
        BlueWinningMC.SetText(LosingText);
    }
}

// Update the roundtime clock
float secondsLeft = penv->GetSecondsLeftInRound();
unsigned sec = ((unsigned)secondsLeft % 60);
if (sec != State.Sec)
{
    unsigned min = ((unsigned)secondsLeft / 60);
    String timeLeft;
    Format(timeLeft, "{0}:{1:0.2}", min, sec);
    RoundTimeMC.SetText(timeLeft);
    State.Sec = sec;
}
}

```

```

                                Scaleform::String      GFx::Value::SetText()      scoreRed  scoreBlue
teamRed  teamBlue      X      GFx::Value::SetDisplayInfo()
0%

blueWinning/redWinning
GFx::Value::SetText()      HUDView      "Winning"      "Losing"
                        textField      UpdateView()
                        GFx::Value::SetText()

roundTime textField

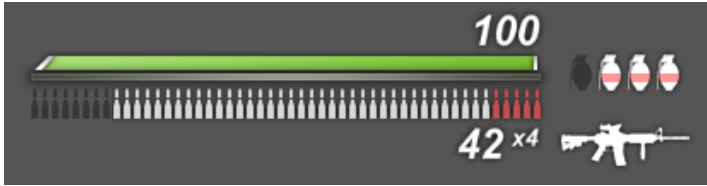
```

4.2 Player Stats Ammo

```

playerStats      HUDKit.fla      1      playerStats
(healthN, health)      (weapon) HUD      (reload)

```



6 playerStats

clipN ammoN ammo ammoBG grenade grenadeBG

clipN ammoN

ammo/healthVehicle

ammo ammoBG

51 Keyframes

GfX::Value::GotoAndStop()

- *ammo*
- *ammoBG*

grenade/armor ammo

grenade grenadeBG

5

1 4

5

[GfX::Value::GotoAndStop\(\)](#)

- *grenade*
- *grenadeBG*

playerStats

```
void FxHUDView::UpdatePlayerStats(FxHUDEnvironment *penv)
{
    FxHUDPlayer* pplayer = penv->GetHUDPlayer();

    // Load latest player info.
    unsigned ammoInClip = pplayer->GetNumAmmoInClip();
    unsigned ammoClips = pplayer->GetNumClips();
    unsigned clipSize = pplayer->GetMaxAmmoInClip();
    unsigned grenades = pplayer->GetNumGrenades();

    unsigned ammoFrames = 51;
    unsigned grenadeFrames = 5;

    String text;

    if(pplayer->GetHealth() != State.Health)
```

```

{
    unsigned health = unsigned(pplayer->GetHealth() * 100);
    Format(text, "{0}", health);
    HealthNMC.SetText(text); // Update the health text field.

    Value::DisplayInfo info;
    info.SetScale(Double(health), 100);
    HealthMC.SetDisplayInfo(info); // Update and scale the health bar.
}

if(ammoInClip != State.AmmoInClip)
{
    Format(text, "{0}", ammoInClip);
    AmmoNMC.SetText(text); // Update the ammo text field.

    // The ammo is divided into two parts: the white bullet icons (AmmoMC)
    // and their respective backgrounds (AmmoBGMC).

    // To update the number of bullets currently in the clip, we use
    // GotoAndStop with AmmoMC and select the frame with the proper number
    // of white bullet icons. There are 51 frames in AmmoMC (defined above
    // in ammoFrames) and the first frame displays every bullet icon.

    // To display X number of bullets, we subtract X from the total number
    // of frames in AmmoMC (51) and GotoAndStop on the result.
    AmmoMC.GotoAndStop(ammoFrames - ammoInClip);
}

if(ammoClips != State.AmmoClips)
{
    Format(text, "x{0}", ammoClips);
    ClipNMC.SetText(text); // Update the remaining ammo clips text field.
}

if(grenades != State.Grenades)
{
    // Grenades are setup similar to Ammo. The first frame for the
    // Grenades movieClip shows 4 white grenade icons. The second frame
    // shows 3.

    // To display X number of grenades, we subtract X from the total number
    // of frames in GrenadeMC (5) and GotoAndStop on the result.
    // Note: GrenadeMC actually contains 10 frames but 6-10 are unused by
    // this implementation.
    GrenadeMC.GotoAndStop(grenadeFrames - grenades);
}

if (pplayer->GetWeaponType() != State.weaponType)

```

```

{
    Change the weapon icon if the player has changed weapons.
    unsigned weaponFrame = GetWeaponFrame(pplayer->GetWeaponType());
    WeaponMC.GotoAndStop(weaponFrame);

    // Update the ammo background icons based on clipSize.
    if (clipSize != State.AmmoClipSize)
        AmmoBGMC.GotoAndStop(ammoFrames - clipSize);
}

// If the ammo in clip is less than 6 bullets, play the "Reload" indicator
// movieClip.
if (ammoInClip <= 5)
{
    // ReloadMC will play until we GotoAndStop on Frame 1, so no need to
    // start the animation more than once.
    if (!bReloadMCVisible)
    {
        bReloadMCVisible = true;
        ReloadMC.GotoAndPlay("on"); // Will cycle back to "on" frame when
                                     it reaches the end and play again.
    }
}

// If the reload indicator is displayed but there are more than six bullets
// hide it and stop the animation by playing frame 1 of the movieClip.
else if (bReloadMCVisible)
{
    ReloadMC.GotoAndStop("1"); // Frame 1 contains stop();
    bReloadMCVisible = false;
}
}

```

```

ammo      1   50                      51   0
ammoMC.GotoAndStop(ammoFrames - ammoInClip)                                HUD

```

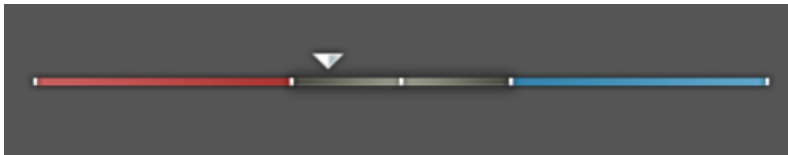
```

ammoBG      ammoBGMC.GotoAndStop()      X
X
(Machine Gun)      50
ammoBG.GotoAndStop(1)      1      50
FxHUDView::UpdateView()      MovieClip

```

4.3

flagMC HUDKit fla 1 (arrow) *flagMC* *flag*
CaptureState



7 *flagMC*

flagMC Classic Tween Flash *flagMC*
4 4 Hidden Visible Fade-In
FadeOut GotoAndStop() Keyframe / 1 5 "On" "Off"

FxHUDView::UpdateEvents() *flagMC* arrow

```
// Flag capture indicators and reticule behavior.
void FxHUDView::UpdateEvents(FxHUDEnvironment *penv)
{
    Value::DisplayInfo info;

    if (penv->IsHUDPlayerCapturingObjective())
    {
        if(!bFlagMCVisible)
        {
            SetVisible(&FlagMC, true); // Set the Flag MovieClip's visibility
                                     // to true
            FlagMC.GotoAndPlay("on"); // Play the fade-in animation once.
            bFlagMCVisible = true;
        }

        // Shift the x-coordinate of the Flag Arrow to show the capture state
        // of the flag the player is currently capturing.
        info.SetX(penv->GetHUDCaptureObjectiveState() * 177);
        FlagArrowMC.SetDisplayInfo(info);
        info.Clear();
    }
    else if (bFlagMCVisible & !penv->IsHUDPlayerCapturingObjective())
    {

```

```

        // If the flag indicator is still visible and the player is
        // no longer capturing an objective, play the fade-out animation
        FlagMC.GotoAndPlay("off");
        bFlagMCVisible = false;
    }
}

```

```

        flagMC.GotoAndPlay("on")    flagMC    Visible
flagMC.GotoAndPlay("off")    flagMC    11-20    20
        1    Flash    ActionScript    C++    1
stop();    1    1    1
        Alpha    0

arrow    Gfx::Value::SetDisplayInfo\(\)
Gfx::Value::DisplayInfo    CaptureState    X

```

4.4

```

Reticule    HUDKit.fla    1    reticule
    left    right    bottom    top

PlayerFire

```



8 reticule

```

        PlayerFire    FxHUDView::UpdateEvents()    FxHUDView::OnEvent()
        ReticuleHeat    ReticuleFiringOffset    FxHUDEvent

// If the player is firing his/her weapon, update the reticule's heat.
// The reticule elements are updated in FxHUDView::UpdateEvents() based on
// the ReticuleHeat and the ReticuleFiringOffset.
case FxHUDEvent::EVT_PlayerFire:
{
    if (ReticuleHeat < 8)
        ReticuleHeat += 2.5f;
}

```

```

    ReticleFiringOffset = 2;
}

FxHUDView::UpdateEvents()
    top    bottom    left    right
    SetDisplayInfo()    X    Y    6    -6
    0, 0    X    Y    SetDisplayInfo()
Gfx::Value::DisplayInfo    DisplayInfo
    Gfx::Value::SetDisplayInfo()    DisplayInfo.Clear()    DisplayInfo

// Shift the XY-coordinates of the reticule elements based on the
// firing duration and rate of fire.
if (ReticleHeat > 0 || ReticleFiringOffset > 0){
    if (ReticleHeat > 1.0f)
        ReticleHeat -= .02f;

    ReticleFiringOffset -= .02f;
    if (ReticleFiringOffset < 0)
        ReticleFiringOffset = 0;

    info.SetY((-6) - (ReticleFiringOffset + ReticleHeat));
    ReticuleMC_Top.SetDisplayInfo(info);
    info.Clear();

    info.SetX((6) + (ReticleFiringOffset + ReticleHeat));
    ReticuleMC_Right.SetDisplayInfo(info);
    info.Clear();

    info.SetX((-6) - (ReticleFiringOffset + ReticleHeat));
    ReticuleMC_Left.SetDisplayInfo(info);
    info.Clear();

    info.SetY((6) + (ReticleFiringOffset + ReticleHeat));
    ReticuleMC_Bottom.SetDisplayInfo(info);
    info.Clear();
}

```

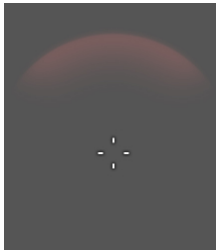
4.5

dirHit HUDKit fla 1 *dirHit* *dirHit*
 tl l bl b br r tr t

Alpha

a)

b)



9 *dirHit*

~1

Flash

Classic Tween Alpha

GFx::Value::GotoAndPlay("on")

```
void FxHUDView::OnEvent( FxHUDEvent* pevent )
{
    switch ( pevent->GetType() )
    {
        // If the Damage Event (Player was hit) is fired, show the appropriate
        // directional hit indicator.
        // We can use GotoAndPlay("on") to play the fade-in animation. It will
        // fade out on its own after ~1 second (this animation is setup on the
        // Flash timeline)
        case FxHUDEvent::EVT_Damage:
        {
            FxHUDDamageEvent* peventDamage = (FxHUDDamageEvent*)pevent;
            float dir = peventDamage->GetDirection();
            if (dir > 0)
            {
                if (dir > 90)
                {
                    if (dir > 135)
                        DirMC_B.GotoAndPlay("on");
                    else
                        DirMC_BR.GotoAndPlay("on");
                }
            }
            else
            {
                if (dir < -90)
                {
                    if (dir < -135)
                        DirMC_L.GotoAndPlay("on");
                    else
                        DirMC_LL.GotoAndPlay("on");
                }
            }
        }
    }
}
```

```

        {
            if (dir > 45)
                DirMC_TR.GotoAndPlay("on");
            else
                DirMC_T.GotoAndPlay("on");
        }
    }
else
{
    if (dir < -90)
    {
        if (dir < -135)
            DirMC_B.GotoAndPlay("on");
        else
            DirMC_BL.GotoAndPlay("on");
    }
    else
    {
        if (dir < -45)
            DirMC_L.GotoAndPlay("on");
        else
            DirMC_TL.GotoAndPlay("on");
    }
}
break;
}
}

```

4.6

expBar HUDKit.fla Scene 1 *expBar*
 expBar *exp* MovieClip

rank HUDKit.fla Scene 1 *rank* *rank* 18 Keyframe
 Keyframe



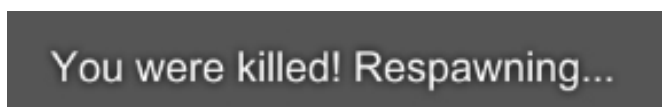
10 *expBar* *rank*

```
void FxHUDView::UpdatePlayerEXP(FxHUDEnvironment *penv)
{
    FxHUDPlayer* pplayer = penv->GetHUDPlayer();
    if (pplayer->GetLevelXP() != State.Exp)
    {
        Value::DisplayInfo info;
        // Update the rank icon. Each rank icon is on a different frame of
        // the rank movieClip.
        RankMC.GotoAndStop(UInt(pplayer->GetRank() + 1));
        info.SetScale(pplayer->GetLevelXP() * 100, 100); //      EXP
                                                         //
        ExpBarMC.SetDisplayInfo(info);
    }
}
```

Gfx::Value::GotoAndStop() X *exp* X + 1 0 *rank*

4.7

centerTextMC HUDKit.fla 1 *centerText*
 centerText



5 *centerTextMC*

<i>centerTextMC</i>	<i>centerText</i>	<i>textField</i>
	<i>centerTextMC</i>	"on5"
5	"on"	3
1	<i>centerTextMC</i>	Alpha 0
		"on" "on5"

```

TextFieldMC.SetText(RespawnText); // Set the text field of the Center Text.
CenterTextMC.GotoAndPlay("on"); // Play the fade-in animation, it will fade on its
                                own after ~3 sec.

```

<i>centerTextMC</i>	<i>textField</i>	<i>GotoAndPlay("on")</i>
Classic Tween	~3	

4.8

<i>popup</i>	HUDKit fla	1	<i>popup</i>
	3		<i>popup</i>
3-5	C++		



6 *popup*

<i>Popup</i>		Classic Tweens	<i>popup</i>
<i>popup</i>	4	<i>actions popupNumber popupText popupBG popupBG</i>	
" "		<i>popupText popupNumber</i>	<i>textField</i>
"on"		<i>GotoAndPlay()</i>	
	1	Alpha	0

```

// If a KillStreak event is fired, display the kill streak pop-up
case FxHUDEvent::EVT_KillStreak:
{
    FxHUDKillStreakEvent* peventKS = (FxHUDKillStreakEvent*)pevent;

    // Format the number of kills
    String text;
    Format(text, "{0}", peventKS->GetSrcKillStreak());
    PTextFieldMC.SetText(text); // PopUp

```

```

        // Play the fade-in animation, it will fade on its own after ~4 sec.
        PopupMC.GotoAndPlay("on");
    }
    break;

```

	FxHUDEvent	KillStreak		
C++		popupText's textField	PTextFieldMC	GotoAndPlay("on")
popup				

4.9

log	HUDKit.fla	Scene 1	log	HUD
				~3



13 log logMessage

logMessage		3
logMessage	C++	htmlText textField

FxHUDMessage::Movieclip*	FxHUDLog::GetUnusedMessageMovieclip()
LogMessage MovieClip	Log

```

FxHUDMessage::Movieclip* FxHUDLog::GetUnusedMessageMovieclip()
{
    if (MessageMCs.IsEmpty())
    {
        // Request a new line in the SWF
        Value logline;
        LogMC.AttachMovie(&logline, "LogMessage", "logMessage"+NumMessages);
        FxHUDMessage::Movieclip* pmc = new FxHUDMessage::Movieclip(logline);
        MessageMCs.PushBack(pmc); // Add new Message MC to list of unused
                                   message movieClips.
    }
}

```

```

    }
    FxHUDMessage::Movieclip* pmc = MessageMCs.GetFirst(); // Use an unused Message
                                                         movieClip.

    pmc->SetVisible(true);
    MessageMCs.Remove(pmc);
    return pmc;
}

```

C++	C++	Level Up	
"You are now a [Rank Icon] [Rank Name]"	[	][	htmlText

```

void FxHUDLog::Process(FxHUDEvent *pevent)
{
    String msg;
    switch (pevent->GetType())
    {
        case FxHUDEvent::EVT_LevelUp:
        {
            FxHUDLevelUpEvent* e = (FxHUDLevelUpEvent*)pevent;
            Format(msg, "You are now a <img src='rank{1}'/> {0}!",
                e->GetNewRankName(), e->GetNewRank());
        }
        break;
    }
}

FxHUDMessage::Movieclip* pmc = GetUnusedMessageMovieclip();
FxHUDMessage* m = new FxHUDMessage(msg, pmc); // 3
Log.PushBack(m);
NumMessages++;

```

HUD Update

```

void FxHUDLog::Update()
{
    SF_ASSERT(LogMC.IsDisplayObject());

    Double rowHeight = 30.f;

    // Tick the message lifetimes
    Double yoff = 0;
    FxHUDMessage* data = Log.GetFirst();
    while (!Log.IsNull(data))
    {
        FxHUDMessage* next = Log.GetNext(data);
        if (data->IsAlive())
        {

```

```

        // Layout mgmt and animation
        Value::DisplayInfo info;
        info.SetY(yoff);
        data->GetMovieclip()->GetRef().SetDisplayInfo(info);
    }
    data = next;
    yoff += rowHeight;
}

// Layout management and animation
Value::DisplayInfo info;
info.SetY(LogOriginalY - (NumMessages * rowHeight));
LogMC.SetDisplayInfo(info);

// Remove dead messages
data = Log.GetFirst();
while (!Log.IsNull(data))
{
    FxHUDMessage* next = Log.GetNext(data);
    if (!data->IsAlive())
    {
        Log.Remove(data);
        NumMessages--;

        FxHUDMessage::Movieclip* pmc = data->GetMovieclip();
        pmc->SetVisible(false);
        MessageMCs.PushBack(pmc);

        delete data;
    }
    data = next;
}

}

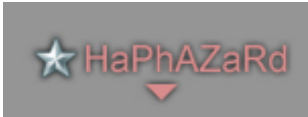
Log                                FxHUDMessage                        0
FxHUDMessage                      Log                                FxHUDMessage MovieClips
    MessageMCs                      FxHUDMessage                    Gfx::Value::DisplayInfo.SetY()
    Gfx::Value::SetDisplayInfo()    MessageMC

```

4.10

billboard_container HUDKit.fla Scene 1 billboard 3D 3D

(Frustum)



7 billboard_enemy

UI

2D

Gfx::Value::SetDisplayInfo()

MovieClip (billboard_friendly, billboard_enemy)

billboard_label_(friendly/enemy)

billboard_label_*

textField billboard_friendly's "Show" Keyframe

billboard_friendly

Classic Tween

"Show"

C++

FxHUDView::BillboardCache::GetUnusedBillboardMovieclip :

```
if (UnusedBillboards.IsEmpty())
{
    Value::DisplayInfo info(false);

    Value billboard, temp;
    String instanceName;
    Format(instanceName, "{0}{1}", BillboardPrefix, BillboardCount++);
    CanvasMC.AttachMovie(&billboard, SymbolLinkage, instanceName);

    BillboardMC* pbb = new BillboardMC();
    pbb->Billboard = billboard;
    billboard.GetMember("label", &temp);
    temp.GetMember("textField", &temp);
    pbb->Textfield = temp;

    pbb->Billboard.SetDisplayInfo(info);
    UnusedBillboards.PushBack(pbb);
}
```

BillboardCache

CanvasMC

BillboardCache

billboard_enemy/billboard_friendly

CanvasMC.AttachMovie()

```
Billboard                                UpdateBillboards()  BeginProcessing()  EndProcessing()
    GetUnusedBillboardMovieclip()
BeginProcessing()  EndProcessing()
```

UpdateBillboards()

```
// Process friendlies
// Friendly billboards will always show names
```

```
BillboardMC* pbb = NULL;
```

```
FriendlyBillboards.BeginProcessing();
```

```
for (unsigned i=0; i < friendlies.GetSize(); i++)
```

```
{
```

```
    info.Clear();
```

```
    if (FriendlyBillboards.GetUnusedBillboardMovieclip(friendlies[i].pPlayer,
                                                         &pbb))
```

```
    {
```

```
        info.SetVisible(true);
```

```
        pbb->Billboard.GotoAndPlay("show");
```

```
        // Load player info
```

```
        String title;
```

```
        Format(title,
```

```
                "<img src='rank{0}' width='20' height='20' align='baseline'
```

```
                vspace='-7' /> {1}";
```

```
                friendlies[i].pPlayer->GetRank()+1,
```

```
                friendlies[i].pPlayer->GetName());
```

```
        pbb->Textfield.SetTextHTML(title);
```

```
    }
```

```
    info.SetX(friendlies[i].X);
```

```
    info.SetY(friendlies[i].Y);
```

```
    info.SetScale(friendlies[i].Scale, friendlies[i].Scale);
```

```
    pbb->Billboard.SetDisplayInfo(info);
```

```
}
```

```
FriendlyBillboards.EndProcessing();
```

```
UpdateBillboards()
```

```
GotoAndPlay("show")
```

```
GFx::Value::SetDisplayInfo()
```

```
MovieClip's X Y
```

```
GFx::Value::SetText()
```

```

textField      htmlText
src='rank{0}' HTML
>GetRank( )+1)
    {1}

```

img

```

+ 1 (friendlies[i].pPlayer-
      vspace
(friendlies[i].pPlayer->GetName( ))

```

5

(Minimap) C++

Apps\Kits\HUD

- **FxMinimap.h**
- **FxMinimapView.h/.cpp**

5.1.1

- **FxMinimapIconType**
- **FxMinimapIcon**
- **FxMinimapIconCache**

ID

/

/

MinimapIconFactoryBase

MovieClips
Update()

textField

o *PlayerIcon*

o *FlagIcon*

o *Objectivelcon*

o *WaypointIcon*

- **FxMinimapView**

UpdateView()

PlayerIcon::Update()

FxMinimapView.cpp

116

Direct

Access API

MovieClip

MovieClip

SetDisplayInfo

Alpha

SetMember()

GFx::Value::SetMember

GFx::Movie::SetVariable

```
virtual void Update(FxMinimapView* pview, FxMinimapEntity* pentity,
                    float distSquared)
```

```
{
```

```
    SF_UNUSED(distSquared);
```

```
    bool hasIcon = (pentity->GetIconRef() != NULL);
```

```
    PointF pt = pentity->GetPhysicalPosition();
```

```
    bool showDir = pview->IsShowingPlayerDir();
```

```
// If entity did not have an icon attached before, then wait for the
```

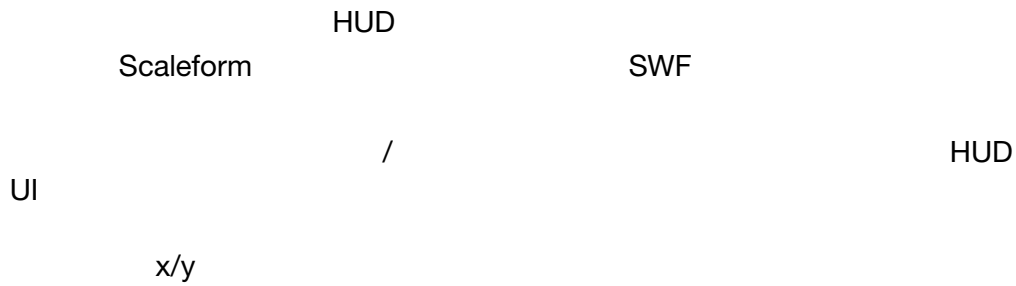
```

// next update to set the state data. picon is most probably not initialized
// (the movie needs to advance after creation)
if (hasIcon && (State != (int)showDir))
{
    // state 0: no arrow, state 1: show arrow
    Value frame(Double(showDir ? 2.0 : 1.0));
    MovieClip.Invoke("gotoAndStop", NULL, &frame, 1);
    State = (int)showDir;
}

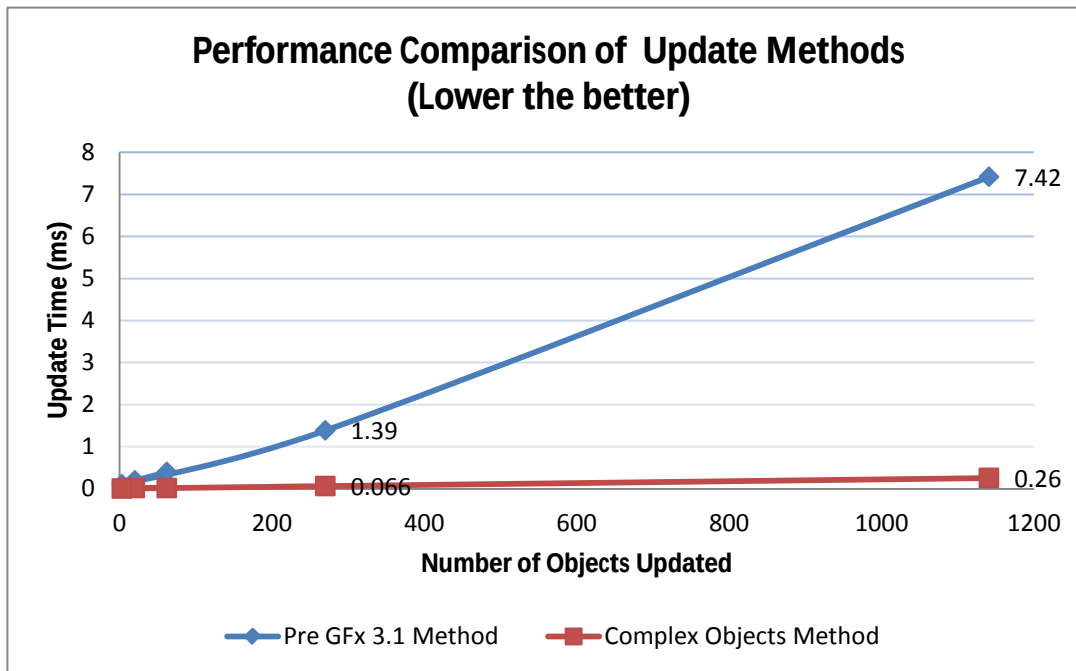
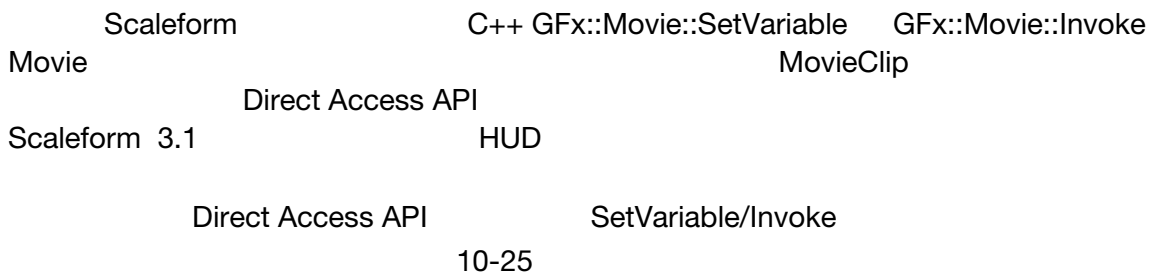
pt = pview->GetIconTransform().Transform(pt);
Value::DisplayInfo info;
if (State)
{
    info.SetRotation(pview->IsCompassLocked() ?
        (pentity->GetDirection() + 90) : (pentity->GetDirection() -
            pview->GetPlayerDirection()));
}
info.SetPosition(pt.x, pt.y);
MovieClip.SetDisplayInfo(info);
}

```

6



6.1



8 Direct Access API SetVariable/Invoke

Direct Access
– HUD UI

Scaleform

Flash HUD

HUD

Windows Vista
MacBook Pro

1 Scaleform HUD

1
– HUD UI

1

6.2

HUD

GFxExport

1.

GFX

.GFX

1858kb

- HUDKit.gfx - 33kb
- Minimap.gfx - 50kb
- fonts_en.gfx - 46kb
- gfxfontlib.gfx - 100kb

- HUDKit.gfx
 - o UI - 603kb
 - o - 72kb
 - o - 51kb
 - HUD 43kb
- Minimap.gfx
 - o - 644kb
- Map.jpg - 259kb